Zero-momentum global dynamics and three-dimensional traveling waves for the mass-critical biharmonic NLS

Research continuation with analytic proofs and frozen interval certificates

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Abstract

This continuation develops three results for the focusing mass-critical biharmonic nonlinear Schrödinger equation. First, a finite-rank, spatially localizable modification of the linearized energy permits outgoing excited-wave gluing with arbitrarily small nonzero separation speed. Applied to the previously certified four-dimensional negative-energy branch, it gives bounded forward-global solutions with exactly zero momentum, finite variance, and negative energy. Second, whole-space interval estimates complete the three-dimensional profile, negative inverse pairing, and all symmetry-reduced angular channels; an implicit-function argument yields negative-energy traveling waves in dimension three. Third, a purely analytic interaction-virial and mass-transport argument replaces a uniform kinetic bound by its first time average and forces separation on entire time intervals. Uniform mass localization instead forces quantitative averaged kinetic grow-up. All numerical conclusions refer to explicit frozen data, not to plots or finite-box eigenvalues alone.

Status and scope. This is an internally checked research manuscript with reproducibility files, not an externally verified or peer-reviewed result. The four-dimensional application retains the explicit companion profile certificates as an input; the three-dimensional certificates are completed here. The new proofs are stated with their actual hypotheses. Priority for the finite-rank gluing mechanism has not been established. A conflict with a published cylindrical blowup statement is addressed in Section 8; identifying a gap in that proof alone would not establish our construction.

1 Equation, conventions, and principal statements

We use

$$\mathrm{i}u_t = \Delta^2 u - |u|^{8/N}u, \qquad u(t) \in H^2(\mathbb{R}^N), \tag{1}$$

and the conserved quantities

$$M(u) = \int |u|^2, \quad E(u) = \frac{1}{2}\,\|\Delta u\|_2^2 - \frac{1}{2+8/N}\,\|u\|_{2+8/N}^{2+8/N}, \quad P(u) = \operatorname{Im}\int \bar{u}\nabla u.$$

Put $A = \Delta^2 + 1$ and $\|h\|_Z^2 = \|\Delta h\|_2^2 + \|h\|_2^2$. A wave $u(t,x) = e^{\mathrm{i}t}U_v(x - vte_1)$ solves (1) precisely when

$$AU_v + \mathrm{i}v\partial_1 U_v - |U_v|^{8/N}U_v = 0. \tag{2}$$

For an admissibly localized mass-critical profile, variation under $U(x) \mapsto \lambda^{N/2}U(\lambda x)$ in $E + M/2 - vP_1/2$ gives

$$E(U_v) = \frac{vP_1(U_v)}{8}. \tag{3}$$

There is no Galilean reduction that forces vP_1 to be positive for this fourth-order equation.

Theorem 1.1 (Four-dimensional zero-momentum construction)**.** *Let Q_v be the previously certified smooth exponentially localized negative-energy cubic BNLS profile, for any fixed admissible $0 < v \le 10^{-6}$. There exist $T, \gamma > 0$ and an exact solution such that, with $Q_{-v}(x) = Q_v(-x)$,*

$$R(t,x) = e^{\mathrm{i}t}\big(Q_v(x - vte_1) + Q_{-v}(x + vte_1)\big),$$

$$\|u(t) - R(t)\|_{H^{10}} + \sum_{j=1}^{4} \|x_j(u(t) - R(t))\|_{H^6} \le Ce^{-\gamma t} \qquad (t \ge T). \tag{4}$$

One can take u inversion-even and invariant under transverse $O(3)$. It has

$$P(u) = 0, \qquad E(u) = 2E(Q_v) < 0, \qquad M(u) = 2M(Q_v), \qquad \sup_{t \ge T} \|u(t)\|_{H^{10}} < \infty.$$

Its variance is finite at every finite time and satisfies

$$\int |x|^2 |u(t,x)|^2 \,\mathrm{d}x = 2M(Q_v)v^2t^2 + 2\int |x|^2 |Q_v(x)|^2 \,\mathrm{d}x + O(te^{-\gamma t}). \tag{5}$$

Time translation gives a forward-global solution with initial time zero.

The profile input already proves $E(Q_v) < -18v^2$ on the displayed explicit speed interval. The gluing theorem itself needs neither an exact Morse count nor the final radial Hamiltonian quartet. Starting separation and all constants may deteriorate as $v \downarrow 0$.

Theorem 1.2 (Three-dimensional negative-energy waves)**.** *The frozen three-dimensional rational profile q specified in Section 4 has an exact real radial solution Q with*

$$AQ - |Q|^{8/3}Q = 0, \qquad \|Q - q\|_Z < 4 \times 10^{-6}.$$

On $X = \{h \in H^2(\mathbb{R}^3;\mathbb{C}) : h(-x) = \overline{h(x)}\}$, its real linearization is invertible with norm less than 200 from X^* to X. Moreover,

$$I = \Big\langle \partial_1 Q, (A - |Q|^{8/3})^{-1}\partial_1 Q \Big\rangle < -1280. \tag{6}$$

There is a C^3 branch solving (2) for $|v| \le 10^{-9}$ with

$$U_v = Q - \mathrm{i}v(A - |Q|^{8/3})^{-1}\partial_1 Q + o_Z(v), \qquad P_1(U_v) = 2Iv + o(v).$$

Quantitatively, $\|U_v - Q\|_Z \le 1000|v|$ and $E(U_v) < -180v^2$ for $0 < |v| \le 10^{-9}$. Thus $e^{\mathrm{i}t}U_v(x - vte_1)$ is an exact global H^2 solution with constant H^2 norm.

Thus the proposed unrestricted implication $E < 0 \Rightarrow$ finite-time collapse has counterexamples in both dimensions three and four, subject to the supplied computer-assisted proof chain. The four-dimensional conclusion survives adding zero momentum and finite variance. This does not classify negative-energy data or settle every fixed dimension separately.

2 A compact finite-rank energy correction

This is the central analytic mechanism in Theorem 1.1. Work over real Hilbert spaces and fix a smooth exponentially localized cubic profile in $\mathbb{R}^4$ with $0 < |v| < 1/2$. Let

$$L_v = A + \mathrm{i}v\partial_1 - \mathcal{V}_v, \quad \mathcal{V}_v h = 2|Q_v|^2 h + Q_v^2 \bar{h}, \quad J = -\mathrm{i}, \quad B_v = JL_v.$$

The free symbol $|\xi|^4 + 1 - v\xi_1$ has minimum $1 - 3(|v|/4)^{4/3} > 0$. The decaying potential is compact from H^2 to H^{-2}, so L_v has finite negative index, finite kernel, and a coercive positive complement. The graph domains are $D(B_v^k) = H^{2+4k}$. These facts verify the hypotheses for the structural decomposition of Lin–Zeng [1].

Lemma 2.1 (Localizable almost-conserved metric)**.** *For every $\varepsilon > 0$ there is a self-adjoint finite-rank form K_ε with smooth compactly supported kernels such that*

$$c_\varepsilon \|h\|_{H^2}^2 \le q_\varepsilon(h) := \langle (L_v + K_\varepsilon)h, h\rangle \le C_\varepsilon \|h\|_{H^2}^2 .$$

On the invariant center space E^c of B_v,

$$|\langle (K_\varepsilon B_v + B_v^* K_\varepsilon)h, h\rangle| \le \varepsilon q_\varepsilon(h). \tag{7}$$

The inequality is a bounded quadratic-form statement on $E^c \subset H^2$.

Proof. Use the full triangular decomposition $H^2 = \bigoplus_{j=0}^6 X_j$ from [1, Theorem 2.1]. Here $X_0 = \ker L_v$, $X_5 = E^u$, $X_6 = E^s$, all blocks except X_3 are finite-dimensional and lie in every graph domain. On the center $X_0 \oplus \cdots \oplus X_4$, the only triangular arrows are

$$4 \longrightarrow \{2,3,1,0\}, \qquad \{2,3\} \longrightarrow \{1,0\}, \qquad 1 \longrightarrow 0.$$

The diagonal operators on blocks $2, 3$ are skew in the coercive metrics $-L_2, L_3$; those on $1, 4$ are finite matrices with imaginary spectrum. Rescaling their Jordan chains gives positive metrics G_1, G_4 with arbitrarily small symmetric generator parts. Set

$$q_\delta = a_0|x_0|^2 + a_1 G_1[x_1, x_1] - L_2[x_2, x_2] + L_3[x_3, x_3] + a_4 G_4[x_4, x_4].$$

Take a_4 large, then a_1 small, then a_0 smaller. Each off-diagonal norm in these coordinates is its original norm times the square root of the target/source weight ratio. Thus $|q'_\delta| \le \delta q_\delta$. The infinite-dimensional metric remains exactly L_3. Extending positively on the two hyperbolic blocks changes L_v by a finite-rank form only.

The dual functions of this correction are smooth. Coordinates on X_1, X_4 are recovered by pairing against $L_v X_4, L_v X_1$, respectively; coordinates on X_2 use $L_v X_2$, and hyperbolic coordinates use the opposite hyperbolic block. For the kernel, use

$$P_0 = \mathrm{Orth}_{X_0}^{H^2}(I - P_5 - P_6).$$

This accounts for the fact that kernel orthogonality need not extend to hyperbolic blocks. All these duals belong to every Sobolev space, so the correction is a finite sum $\sum c_{jk} \langle f_j, h\rangle \langle f_k, h\rangle$ with smooth Sobolev f_j.

Replace f_j by smooth compact cutoffs $\chi_R f_j$. They converge in each fixed Sobolev norm. Both the form error and its generator-commutator error tend to zero: in a rank-one term move B_v to the smooth dual function, using $B_v^* : H^2 \to H^{-2}$. First fix the small-rate metric, then choose the support radius relative to its coercivity. Positivity and (7) follow. No decay assumption on embedded imaginary eigenmodes is used. □

For the existing branch the correction rank can be bounded by $\dim\ker L_v + 2n^-(L_v) = 5 + 2\cdot 36 = 77$. The argument needs only finiteness; this number is not a hypothesis for gluing.

3 Moving-center gluing and weighted regularity

3.1 The hyperbolic projection and center evolution

Only the finitely many off-axis generalized eigenvectors and their duals must be transported exactly. They are smooth and exponentially localized: the free fourth-order resolvent is invertible off the imaginary spectrum, small exponential conjugation is a relative graph perturbation, and exterior potential terms can be absorbed. Induction treats Jordan chains.

Translate synthesis and analysis maps to the two centers and correct the exponentially small off-diagonal Gram matrix. Write them $Z(t), W(t)$ with $WZ = I$, and let B_0 be the direct sum of the two hyperbolic generators. With

$$S = Q_v(\cdot - vte_1) + Q_{-v}(\cdot + vte_1), \quad \mathcal{A} = J(A - \mathcal{V}_S),$$

the errors in

$$\dot{Z} + ZB_0 - \mathcal{A}Z = E_Z, \qquad \dot{W} + W\mathcal{A} - B_0W = E_W$$

are exponentially small in every fixed graph norm used here. Define

$$\Pi = I - ZW, \qquad \mathcal{C} = \mathcal{A} - ZE_W.$$

Then $w_t = \mathcal{C}w$ preserves $W(t)w(t) = 0$.

Choose a real axial velocity field b equal to v near the right core and $-v$ near the left, with transition length comparable to vt. Set

$$D_b = b\partial_1 + \tfrac{1}{2}\partial_1 b, \qquad \mathcal{H}_\varepsilon = A - \mathcal{V}_S + \mathrm{i}D_b + K_\varepsilon^+(t) + K_\varepsilon^-(t).$$

Its coefficients satisfy $\|\partial_x^j b\|_\infty = O_v(t^{-j})$ and $\|b_t\|_\infty = O_v(t^{-1})$. A square partition $\chi_+^2 + \chi_-^2 + \chi_0^2 = 1$ has core cutoffs equal to one on the compact correction supports. Fourth-order IMS and the free exterior positivity give, for sufficiently large t,

$$Q_t(w) := \langle \mathcal{H}_\varepsilon w, w\rangle \asymp_\varepsilon \|w\|_{H^2}^2 , \tag{8}$$

$$q_\varepsilon^+(\chi_+ w) + q_\varepsilon^-(\chi_- w) \le 2Q_t(w). \tag{9}$$

The constant 2 in (9), after increasing the starting time, is essential: replacing local energies by a norm would insert an uncontrolled condition number in front of ε.

For $\mathcal{H}_0 = A - \mathcal{V}_S + \mathrm{i}D_b$, direct real-form differentiation gives

$$\dot{\mathcal{H}}_0 + \mathcal{H}_0\mathcal{A} + \mathcal{A}^*\mathcal{H}_0 = \mathrm{i}D_{b_t} + [D_b, \Delta^2] - ((\mathcal{V}_S)_t + b\partial_1\mathcal{V}_S). \tag{10}$$

Transport cancels the last term in the cores; tails are exponentially small. Integrating the commutator as a quadratic form bounds it by $C_v t^{-1}\|w\|_{H^2}^2$. The translation derivative of each $K_\varepsilon^\pm$ cancels the matching transport. The remaining compact form is exactly the single-wave form in (7), up to exponential errors.

If $Ww = 0$, the local vectors $\chi_\pm w$ have only exponentially small single-wave hyperbolic coordinates, since those duals decay exponentially. Project them onto the corresponding centers and use (9). Consequently

$$|Q_t'(w)| \le (2\varepsilon + C_\varepsilon/t + C_\varepsilon e^{-\kappa t})Q_t(w),$$

and the center propagator satisfies

$$\|U_C(t,s)\|_{H^2\to H^2} \le C_\varepsilon e^{\varepsilon|t-s|}\max(t/s, s/t)^{m_\varepsilon}. \tag{11}$$

This provides any prescribed small exponential rate, even for arbitrarily small fixed v.

3.2 Graph norms and one spatial moment

Let $\mathcal{D} = \Pi\mathcal{A}\Pi$. The difference $\mathcal{D} - \mathcal{A}$ is finite rank and smoothing. On the moving center,

$$\|w\|_{H^6} \asymp \|w\|_{H^2} + \|\mathcal{D}w\|_{H^2}.$$

For $z = \mathcal{D}w$,

$$z_t = \mathcal{C}z + \mathcal{F}w, \qquad \mathcal{F} = \dot{\mathcal{D}} + [\mathcal{D}, \mathcal{C}].$$

Here $\mathcal{F}$ is an order-zero smooth multiplier plus smooth finite-rank terms. It is bounded on each fixed Sobolev space, and $\mathcal{F}w$ belongs to the moving center. Duhamel gives (11) in H^6, with an additional polynomial factor. Iterating

$$(\mathcal{D}^k)_t + [\mathcal{D}^k, \mathcal{C}] = \sum_{j=0}^{k-1} \mathcal{D}^j \mathcal{F} \mathcal{D}^{k-1-j}$$

gives the same small exponential rate in H^{2+4k}.

For position weights set $z_j = \Pi x_j w$ and $B = \dot{\Pi} + \Pi\mathcal{C} - \mathcal{C}\Pi$. Since $B\Pi = 0$,

$$(z_j)_t - \mathcal{C} z_j = B(I - \Pi) x_j w + \Pi[x_j, \mathcal{C}] w. \tag{12}$$

The raw hyperbolic constraints vanish individually. A translated dual therefore satisfies

$$\langle \psi(\cdot - c_k(t)), x_j w \rangle = \langle (x_j - c_{k,j}(t)) \psi(\cdot - c_k(t)), w \rangle .$$

This removes the growing center displacement. The right side of (12) is bounded from H^{10} to H^6, uniformly for large t: the free position commutator has order three, same-core finite-rank centers cancel, and cross-core polynomial factors multiply exponential errors. Thus the center bound extends to

$$Y = H^{10} \cap \{h : x_j h \in H^6,\ 1 \le j \le 4\},$$

again with only additional polynomial factors. On finite hyperbolic coordinates, the Y norm is comparable to $(1+t)|a|$, so transport ratios also cost only a polynomial in $|t-s|$.

3.3 The exact nonlinear solution

The interaction defect is $O_Y(e^{-\kappa t})$, after decreasing κ to absorb position factors. Let $\alpha > 0$ be a hyperbolic decay rate, omitting it if the hyperbolic space is empty. Choose

$$0 < \gamma < \min(\kappa, \alpha), \qquad 0 < \varepsilon < \gamma/2,$$

and only then choose the compact metric and a sufficiently large T. In the exponentially weighted norm $\sup_{t \ge T} e^{\gamma(t-T)} \|h\|_Y$, the center integral backward from infinity converges because

$$\int_0^\infty (1+r)^{m_\varepsilon} e^{-(\gamma-\varepsilon)r} \, \mathrm{d}r < \infty.$$

Stable coordinates are integrated forward and unstable coordinates backward. The remaining exponential couplings are absorbed by a Neumann series. The cubic nonlinear remainder is quadratic and Lipschitz on Y; each position factor can be assigned to a correction factor, for example

$$\left\| x_j (S|h|^2) \right\|_{H^6} \le C \|S\|_{H^6} \|h\|_{H^{10}} \|x_j h\|_{H^6} .$$

The uniform unweighted norms of S suffice. A contraction for sufficiently large T proves (4).

Reflection can be imposed throughout. Compact transverse symmetry can be imposed by averaging the bounded Green right inverse over $O(3)$ and restricting the contraction to invariant functions. This averages the right inverse, not the finite-rank metric. Conservation and asymptotic additivity yield the exact M, E, P identities. The profile density is inversion-even, with zero intrinsic centroid; weighted convergence proves (5).

For sufficiently small v, the one-wave stable dimension is exactly 31, using the additional certificate in Section 6. Prescribing small reflected stable coordinates at T gives an injective family with 31 real parameters and identical M, E, P. This family retains inversion symmetry and finite variance. Its full parameter count is not asserted after additionally imposing transverse $O(3)$ symmetry, and it is not an open set of bounded data.

4 Whole-space three-dimensional certification

The reference file is `round4_n3_mixed_exact.json`, with SHA-256

`5d2a86dc991fe960e7036ca651634c556ded9da4aa7d1ae357ab5f4f656d526b`.

It specifies rational compactly supported radial polynomials on $[0, 54]$. Mixed auxiliary fields reduce fourth-order residuals to L^2 polynomial integrals. The existing verified bounds are

$$\left\| Aq - |q|^{8/3}q \right\|_{Z^*} < 1.3 \times 10^{-7}, \qquad \|q\|_Z < 137, \qquad \|q\|_\infty < 6.3.$$

The embedding $\|h\|_\infty \le (4\sqrt{2}\pi)^{-1/2} \|h\|_Z < \|h\|_Z /4$ is used below.

4.1 Radial inverse and exact nonlinear profile

Let $V = (11/3)|q|^{8/3}$. Under the unitary radial map $f \mapsto \sqrt{4\pi}\, rf$, A becomes H^2+1 on the half-line, where $H = -\partial_r^2$ is Dirichlet at zero. Compare $\sqrt{V}(H^2+1)^{-1}\sqrt{V}$ with its Dirichlet-box resolvent at $R = 64$. For $\kappa = (1+\mathrm{i})/\sqrt{2}$, the second-order kernel difference is the rank-one expression

$$\frac{e^{-\kappa R}}{\kappa \sinh(\kappa R)} \sinh(\kappa r) \sinh(\kappa s).$$

Taking its imaginary part and bounding the weighted norm gives the whole-space/box error

$$\frac{2}{e^{\sqrt{2}R} - 1} \int_0^{54} V(r) e^{\sqrt{2}r}\, \mathrm{d}r < 6.162 \times 10^{-31}.$$

The exact potential is linearly interpolated on 65536 intervals. Polynomial bounds on V'' give interpolation error below 0.000437020. Exact cosine moments of this interpolant are evaluated by a ball-arithmetic radix-two FFT and the piecewise-linear sinc-squared formula.

Use the first 640 sine modes. Truncate the *positive resolvent series before multiplication by* $\sqrt{V}$; its remainder norm is at most $\|V\|_\infty /[1 + ((641\pi/64)^4)] < 0.000486654$. This avoids dropping an uncontrolled off-diagonal block of a preconditioned potential matrix. A floating eigensystem is only a frozen witness: ball bounds on its Gram error and eigensystem residual certify a finite-matrix spectral error below 3.227×10^{-14}.

Combining all errors proves

$$\operatorname{dist}(1, \operatorname{spec}(A^{-1/2} V A^{-1/2})) > 0.04230627284 > 1/25.$$

The Newton map with radial inverse norm < 25 is invariant and contracting on $\|h\|_Z \le 4 \times 10^{-6}$, since

$$25(1.3 \times 10^{-7}) + \frac{25 \cdot 211}{8} (4 \times 10^{-6})^2 < 4 \times 10^{-6}$$

and its Lipschitz bound is below 0.006. This proves exact radial existence.

4.2 All angular channels, with no spectral truncation assumption

The real part of X has even angular degree and the imaginary part odd degree. For an angular harmonic Y_ℓ, write $\phi = r^\ell Y_\ell f(r)$ and $H_0 = r^\ell Y_\ell h(r)$. The reduced radial Laplacian is $D_D f = f'' + (D-1)f'/r$, $D = 2\ell + 3$. For each exact rational $\mu > 0$, set

$$d_1 = \Delta\phi - H_0, \qquad d_2 = \Delta H_0 + \phi - \mu^{-1}\beta|q|^{8/3}\phi.$$

Then $K = A^{-1}\beta|q|^{8/3}$, self-adjoint in Z, satisfies

$$\|K\phi - \mu\phi\|_Z \le \mu(\|d_1\|_2^2 + \|d_2\|_2^2)^{1/2}.$$

All radial factors have nonnegative integer powers at the origin. Noninteger powers of q are enclosed by zero-safe Taylor estimates. Norm lower bounds and endpoint matching are verified from the frozen polynomials.

Each of the disjoint intervals $[\mu_j - 10^{-3}, \mu_j + 10^{-3}]$ contains a genuine eigenvalue. An independent whole-space Green-kernel Hilbert–Schmidt bound τ gives

$$\mu_{\text{omitted}} \le \left(\tau - \sum_j (\mu_j - 10^{-3})^2 \right)^{1/2}.$$

Thus the residual certificates identify actual spectrum and the remaining square budget controls *all* omitted spectrum. No orthogonality of numerical trials is assumed.

Sector	Certified reference-profile information	Completion
Real even $\ell = 0$	radial gap > 0.04230	exact-profile transfer
Real even $\ell = 2, 4, 6, 8, 10, 12, 14$	gap > 0.0645 in each sector	mixed residuals + HS
Real even $\ell \ge 16$	$\mathrm{HS}^2 < 0.702$ at $\ell = 16$	angular monotonicity
Imaginary odd $\ell = 1$	gap > 0.006	four BS intervals + HS
Imaginary odd $\ell = 3$	gap > 0.041	mixed residuals + HS
Imaginary odd $\ell \ge 5$	strict positivity at $\ell = 5$	angular monotonicity

The potential transfer is at most $(8/3)(6.3 + 10^{-6})^{5/3} 10^{-6}$ for L_-, and $11/3$ times this for L_+. Every exact-profile channel retains gap $> 1/200$. Monotonicity is proved on the effective-dimensional radial forms; it is used only after a strictly positive channel, not across a negative channel. Taking the Hilbert direct sum proves the full inverse in Theorem 1.2.

4.3 The inverse pairing and the traveling branch

An independently frozen dipole $Z_0 = x_1 f(r)$ satisfies

$$\|Z_0\|_Z < 808.262, \quad \|L_-(q) Z_0 - \partial_1 q\|_{Z^*} < 0.000182258, \quad \langle \partial_1 q, Z_0 \rangle < -1323.3970.$$

For $g = \partial_1 Q$, $L = L_-(Q)$, $r = g - L Z_0$, the exact identity

$$\left\langle g, L^{-1} g \right\rangle = 2 \langle g, Z_0 \rangle - \langle L Z_0, Z_0 \rangle + \left\langle r, L^{-1} r \right\rangle \tag{13}$$

makes the inverse error quadratic in the response residual. The certified potential transfer gives $\|r\|_{Z^*} < 0.046504$ and $\|L^{-1}\| < 200$. Substitution, including the $Q - q$ derivative error, gives $I < -1285.375835741006$, hence (6).

The map $(v, U) \mapsto AU + \mathrm{i} v \partial_1 U - |U|^{8/3} U$ is C^3 from $\mathbb{R} \times X$ to X^*. The implicit-function theorem gives the stated expansion. The following direct contraction supplies an explicit interval. Write $z = L_-^{-1} \partial_1 Q$, so that $\|z\|_Z < 818$, and let $N(w) = g(Q + w) - g(Q) - Dg(Q) w$. On $\|w\|_Z \le 1000 |v|$, $|v| \le 10^{-9}$, the real two-component derivative obeys

$$\left\| D^2 g(\zeta) \right\| \le \frac{88}{9} |\zeta|^{5/3} < 211.$$

The embedding and $\|\partial_1\|_{Z \to Z^*} < 3/5$ imply

$$\|N(w)\|_{Z^*} \le \frac{211}{8} \|w\|_Z^2, \qquad \mathrm{Lip}(N) \le \frac{211}{4} 1000 |v|.$$

Because $L = L_+ \oplus L_-$ is real linear, the correct fixed-point equation keeps multiplication by i inside its inverse:

$$w = -\mathrm{i} v z + L^{-1} N(w) - v L^{-1} (\mathrm{i} \partial_1 w).$$

Its image has norm at most $818|v| + 5{,}275{,}120{,}000v^2 < 1000|v|$, and its Lipschitz constant is at most 0.01055012. Moreover

$$\|w + \mathrm{i}vz\|_Z \le 5{,}275{,}120{,}000v^2, \qquad |P_1(U_v) - 2Iv| < 1.1 \times 10^{12} v^2.$$

The latter follows from $\|\nabla f\|_2 \le \|f\|_Z / \sqrt{2}$, $\|Q\|_Z < 137$ and the quadratic momentum expansion. The cutoff dilation proof of (3) applies to this elliptic H^2 profile: elliptic regularity gives H^4, and all cutoff errors vanish by integrable tails. Therefore

$$8E(U_v) \le -2560v^2 + 1.1 \times 10^{12}|v|^3 \le -1460v^2,$$

proving Theorem 1.2. The noninteger nonlinearity is not claimed analytic. In particular this result does not automatically transfer the cubic H^{10} gluing construction to dimension three.

The small speed range is at frequency one. Exact scaling gives

$$u_{\lambda,v}(t,x) = \lambda^{3/2} e^{\mathrm{i}\lambda^4 t} U_v(\lambda x - v\lambda^4 t e_1),$$

with physical speed $\lambda^3 v$ and energy $\lambda^4 E(U_v)$. Hence any prescribed nonzero physical velocity, or any prescribed negative energy, can separately be realized. Independent simultaneous prescription of mass, energy and velocity is not asserted.

5 A sharper four-dimensional profile certificate

The original frozen four-dimensional profile is unchanged. Introduce a rational auxiliary z on its existing mesh by nodal jets

$$z = q'' + 3q'/r, \quad z' = q''' + 3q''/r - 3q'/r^2, \quad z'' = q^3 - q - 3z'/r,$$

and the differentiated formula for z'''. At zero use the smooth radial values, and at the compact endpoint use zero jets. Septic Hermite interpolation gives an admissible global auxiliary field; these are definitions of z, not assertions that q solves the ODE. With

$$d_1 = \Delta q - z, \quad d_2 = \Delta z + q - q^3, \qquad Aq - q^3 = \Delta d_1 + d_2,$$

exact polynomial integration in 256-bit balls yields

$$\|d_1\|_2 < 1.9206685514 \times 10^{-8}, \quad \|d_2\|_2 < 4.1805837651 \times 10^{-8}, \quad \left\|Aq - q^3\right\|_{Z^*} < 4.601 \times 10^{-8}.$$

The original radial Ritz lower bounds $\mu_1 \ge 5.395$, $\mu_2 > 2.999$ and full HS-square bound < 38.80 imply $\mu_3 < 0.84$, hence inverse norm $< 25/4$. With $\rho = 3 \times 10^{-7}$,

$$\frac{25}{4}\left(4.601 \times 10^{-8} + \frac{(312+\rho)\rho^2}{256}\right) < \rho,$$

and the contraction constant is below 10^{-5}. Uniqueness on the old 0.006 ball identifies the same exact Q, now with

$$\|Q - q\|_Z < 3 \times 10^{-7}, \qquad \left\|A^{-1/2}(Q^2 - q^2)A^{-1/2}\right\| < 2.438 \times 10^{-7}. \tag{14}$$

This strengthens subsequent operator transfers by a factor of 20000 in profile radius without fitting a new numerical profile.

6 A certified radial quartet and the exact unstable dimension

This additional result is independent of the new gluing mechanism. At the exact four-dimensional stationary profile, the radial Hamiltonian generator

$$\mathcal{A}_Q = \begin{pmatrix} 0 & L_- \\ -L_+ & 0 \end{pmatrix}$$

has exactly one eigenvalue, counted with algebraic multiplicity, in

$$1.5 < \operatorname{Re}\lambda < 4.5, \qquad 4.5 < \operatorname{Im}\lambda < 7.5. \tag{15}$$

It is therefore algebraically simple in this sector, and the Hamiltonian symmetries give a simple quartet. Its numerical approximation was used to choose test functions only; the certified location is the whole rectangle (15).

6.1 Finite block and infinite positive complement

Quotient the phase kernel $(0, Q)$ using the Z-orthogonal complement. Let $\widehat{A}$ be the induced group generator and H the bounded energy form operator there. The previous radial Morse certificate gives exactly three negative directions and invertibility of H, with $\|H\| < 6$. Choose seven smooth compactly supported trial vectors: four real quartet directions and three critical zero-chain directions. Freeze their endpoint jets and a change of basis as exact dyadic rationals. Put $W : \mathbb{R}^7 \to D(\widehat{A})$ and let B_0 be the explicit conjugate of two real rotation-growth blocks and a length-three nilpotent block.

The rigorous integrations certify that $G = W^*HW$ has all three negative directions. Its energy-orthogonal complement C is positive and coercive. To see coercivity, let $m = \left\|H^{-1}\right\|^{-1}$ and $P_C = I - WG^{-1}W^*H$. Extending a functional on C by P_C and applying H^{-1} gives the compressed inverse bound $\left\|H_C^{-1}\right\| \le \|P_C\| / m$.

The off-diagonal blocks of $\widehat{A}$ in $\operatorname{Ran} W \oplus C$ are bounded: use $W^*H\widehat{A} = -(\widehat{A}W)^*H$ and $W\mathbb{R}^7 \subset D(\widehat{A})$. Removing them preserves the group-generator property. The complementary group conserves its positive energy, so its resolvent in that norm is at most $1/\operatorname{Re}\lambda$.

Write $g \ge \|G^{-1}\|$, $w \ge \|W\|$, $M \ge \|H\|$, $q_c = 1 + w^2 gM$, and $\epsilon \ge \left\|\widehat{A}W - WB_0\right\|$. If B is the exact compressed finite matrix, the Schur correction has norm at most

$$gMq_c^2\epsilon^2 / \operatorname{Re}\lambda.$$

Thus on a contour Γ with $\operatorname{Re}\lambda \ge \alpha_\Gamma > 0$, the condition

$$k\left(\delta_B + \frac{gMq_c^2\epsilon^2}{\alpha_\Gamma}\right) < 1, \quad k \ge \sup_\Gamma \left\|(B_0 - \lambda)^{-1}\right\|, \quad \delta_B \ge \|B - B_0\|, \tag{16}$$

preserves the finite Schur determinant's zero count. Analytic Schur factorization transfers algebraic multiplicity to the full generator. This controls the entire infinite complement without discretizing it.

6.2 Exact-profile transfer and the verified margin

The frozen trial functions are C^5 compact radial polynomials of degree eleven. Origin odd jets through order five and endpoint jets through order five are checked exactly. At q, exact polynomial integrations give the Gram inertia, the signed compressed matrix, and the whole-space H^2 strong residual. In particular $\|B_q - B_0\| < 9.706 \times 10^{-9}$; estimating this quantity only by the strong residual would lose decisive cancellation.

For $a = Q - q$, (14), the original strong residual $\|Aq - q^3\|_2 < 0.000242$, and a Fourier interpolation estimate give

$$\|Aa\|_2 < 0.0003, \qquad \|a\|_\infty < 8.4 \times 10^{-7}.$$

Explicit product estimates for $(Q^2 - q^2)U = (2qU)a + Ua^2$ transfer every strong residual to the exact profile; $\|\nabla a\|_4^2 \le \|a\|_Z \|Aa\|_2 /16$ controls the quadratic derivative term. The phase quotient error is bounded by $\|Q - q\|_Z$ times the corresponding fixed coordinate row norm. Gram and compressed-matrix transfers give the following conservative bounds:

Quantity	Verified upper bound
ϵ	0.002991362
g	5.858206
w	1.000001876
q_c	36.149366
δ_B	0.000877055
k	1.330629

Each contour side is covered by 160 closed interval segments, with rigorous complex matrix inversion over each entire segment. For $M = 6$ and $\alpha_\Gamma = 1.5$, the left side of (16) is less than 0.365771. The reference block has exactly one simple eigenvalue inside; therefore so does the continuum radial generator.

These two real unstable dimensions persist under sufficiently small traveling speed. Together with the previously established 29 nonradial unstable dimensions and the moving-wave Hamiltonian index upper bound 31, they give

$$\dim_{\mathbb{R}} E^u(Q_v) = \dim_{\mathbb{R}} E^s(Q_v) = 31$$

for sufficiently small nonzero v. This certification does not give an explicit velocity threshold for that exact dimension statement.

7 Time-averaged kinetic control and intrinsic separation

This section is independent of every profile and interval certificate. Let a forward-global strong H^2 solution of (1) have $M > 0$, $E < 0$, $P = 0$. Put

$$K(t) = \|\Delta u(t)\|_2, \quad h(t) = 1 + K(t)^2, \quad \Lambda = 16M|E|, \quad \theta = \min(4/N, 1/4),$$

$$d_R(t) = M^{-2} \iint_{|x-y|>R} |u(t,x)|^2 |u(t,y)|^2 \, \mathrm{d}x \, \mathrm{d}y.$$

7.1 A finite-window virial inequality

Choose a smooth radial ϕ_R with quadratic core, bounded gradient $O(R)$, constant exterior and $D^2\phi_R \le I$. Set

$$I_R = 2 \iint |u(y)|^2 \nabla\phi_R(x-y) \cdot \mathrm{Im}(\bar{u}(x)\nabla u(x)) \, \mathrm{d}x \, \mathrm{d}y.$$

The localized fourth-order identity of [2] and its interaction version, as used in [3], give the following bounds with the kinetic dependence retained:

$$|I_R| \le ARh^{1/4}, \qquad I_R' \le -\Lambda + BR^{-2}h^{1/2} + Chd_R^\theta. \tag{17}$$

Here A, B, C depend only on dimension, mass and cutoffs, not on a temporal supremum of K.

For clarity, the closure uses mass tails alone. With $G = M^{1/4}K^{1/2}$ and $H_R = K + cG/R + c\sqrt{M}/R^2$, localized Gagliardo–Nirenberg bounds the nonlinear tail by $C_M H_R^2 d_R^{\min(4/N,1)}$. The cross-flux tail is bounded by $C_M K^{3/2} H_R^{1/2} d_R^{1/4}$. These are at most Chd_R^θ. The leading Hessian term has a favorable upper sign, so no smallness of curvature tails is needed. The derivative falling on the second mass density is retained; its uncut contribution is $-8P \cdot W$, $W = \operatorname{Im} \int \Delta u \nabla \bar{u}$, and vanishes here. Remaining cutoff errors are $BR^{-2}h^{1/2}$. Smoothing on compact time intervals justifies the integrated identity in H^2.

On $J = [S, S+T]$ define $H_J = T^{-1} \int_J h$. Choose a in the first quarter and b in the last quarter with $h(a), h(b) \le 4H_J$. Then $b - a \ge T/2$. Integrating (17) and using Jensen gives

$$\boxed{\frac{\Lambda}{2} \le A_0 \frac{RH_J^{1/4}}{T} + B\frac{H_J^{1/2}}{R^2} + \frac{C}{T}\int_J h(t) d_R(t)^\theta \, \mathrm{d}t.} \tag{18}$$

This holds on every finite window inside the lifespan, without a global kinetic hypothesis.

7.2 Mass transport closes the first-moment endpoint

Choose smooth ψ equal to zero on $|z| \le 1/2$ and one on $|z| \ge 1$, with $0 \le \psi \le 1$. The smoothed pair tail

$$F_R = M^{-2} \iint \psi((x-y)/R)|u(x)|^2|u(y)|^2 \, \mathrm{d}x \, \mathrm{d}y$$

satisfies $d_R \le F_R \le d_{R/2}$. Testing the density equation with real f gives

$$\int f \partial_t |u|^2 = 2 \operatorname{Im} \int (\Delta f)\bar{u}\Delta u + 4 \operatorname{Im} \int \nabla f \cdot (\Delta u \nabla \bar{u}).$$

It follows that $|F_R'| \le D(R^{-1}K^{3/2} + R^{-2}K)$ and hence

$$|F_R(t) - F_R(s)| \le D\left[R^{-1}\Big(\int_J K^2\Big)^{3/4}|t-s|^{1/4} + R^{-2}\Big(\int_J K^2\Big)^{1/2}|t-s|^{1/2}\right]. \tag{19}$$

The flux exponent $3/2$ is strictly below the kinetic budget exponent 2.

Theorem 7.1 (Separation on an entire time interval)**.** *Suppose for some $H_0, T_* < \infty$,*

$$\sup_{S \ge 0, T \ge T_*} \frac{1}{T}\int_S^{S+T} (1 + K(t)^2) \, \mathrm{d}t \le H_0.$$

There are $c, \delta, \eta, T_0 > 0$ such that every window $[S, S+T]$, $T \ge T_0$, contains an interval of length at least ηT on which, simultaneously for every $a \in \mathbb{R}^N$,

$$\int_{|x-a|>cT} |u(t,x)|^2 \, \mathrm{d}x \ge \delta M. \tag{20}$$

No spatial moment, radiality, higher kinetic time moment or uniform integrability is assumed.

Proof. Set $R = c_0 T$ with c_0 small. The first two errors in (18) are absorbed for large T, giving a positive lower bound on $T^{-1}\int_J h d_R^\theta$. Since $H_J \le H_0$, some $t_0 \in J$ satisfies $d_R(t_0) \ge \delta_0 > 0$. For $|t - t_0| \le \eta T$, (19) is bounded by

$$D(c_0^{-1}H_0^{3/4}\eta^{1/4} + c_0^{-2}H_0^{1/2}T^{-1}\eta^{1/2}).$$

Choose η small to make this at most $\delta_0/2$. Then $d_{R/2} \ge F_R \ge \delta_0/2$ on an interval of length ηT within J. Finally $d_{2r} \le (2/M)\int_{|x-a|>r} |u|^2$ for every center a, which proves (20) after renaming constants. □

There is also an unconditional finite-window alternative: either $H_J > (T/L)^2$, or an interval of length at least $c_\eta H_J^{-4(1+1/\theta)} T$ satisfies

$$d_{(a_0/2)TH_J^{-1/4}} \geq c_\delta H_J^{-1/\theta}.$$

This follows by taking $R = a_0 T H_J^{-1/4}$ in (18) and then applying (19); all constants are independent of the window.

7.3 Localization forces averaged grow-up

Theorem 7.2 (Uniform long-window grow-up)**.** *If the same global solution is uniformly tight in mass modulo translations, namely*

$$\lim_{R\to\infty} \sup_{t\geq 0} \inf_a \frac{1}{M} \int_{|x-a|>R} |u(t,x)|^2 \,\mathrm{d}x = 0,$$

then

$$\lim_{T\to\infty} \inf_{S\geq 0} \frac{1}{T} \int_S^{S+T} K(t)^2 \,\mathrm{d}t = \infty.$$

If the mass tail is uniformly at most DR^{-q}, $0 < q \leq 2$, then, for all sufficiently large T and every $S \geq 0$,

$$\frac{1}{T} \int_S^{S+T} K(t)^2 \,\mathrm{d}t \geq c T^{4q\theta/(4+q\theta)}. \tag{21}$$

Proof. Mass tightness implies $\mu(R) := \sup_t d_R(t) \to 0$. If arbitrarily long windows had bounded H_J, take $R = cT$ in (18). The endpoint term is small for small c, the cutoff term tends to zero, and the tail term is at most $CH_J\mu(cT)^\theta \to 0$, a contradiction.

For a power tail let $\alpha = q\theta \leq 1/2$ and take $R = L_1 H_J^{1/\alpha}$. A sufficiently large fixed L_1 absorbs both the cutoff and tail terms, since $H_J^{1/2-2/\alpha} \leq 1$ and $H_J R^{-\alpha} = L_1^{-\alpha}$. The remaining inequality is $\Lambda/4 \leq CH_J^{1/4+1/\alpha}/T$. Rearrangement and subtraction of the one in H_J prove (21). □

The conclusion is about time averages, not pointwise convergence of $K(t)$ to infinity. Algebraic tails with $q \leq 2$ need not imply finite variance.

8 Relations to prior work and boundaries of the conclusions

The construction combines existing Hamiltonian structure, variational identities and rigorous spectral approximation. Lin–Zeng [1] supplies the triangular decomposition; the deduction here is the compact finite-rank metric with an arbitrarily small center growth rate and its moving-core use. Excited multi-soliton construction has substantial precedent, including the high-speed NLS work of Côte–Le Coz [4]. That result is not itself an arbitrary-small-speed gluing theorem for these fourth-order negative-energy profiles.

The interaction virial also has precedent. Our pure analytic extension changes the closure: low-kinetic endpoint selection and the mass-transport time modulus require only first time averages and yield intervals of separation. The comparison with Komada's compact-orbit argument does not amount to a claim of priority for the interaction functional itself.

There is a material published conflict. Gou's mass-critical cylindrical statement [5, Theorem 1.2], at zero lower-order dispersion, asserts finite-time blowup or infinite-time H^2 grow-up for negative-energy data in a cylindrical class with finite axial moment. Theorem 1.1, if verified as presented, lies in that class and is incompatible with that conclusion. The relevant virial weight is

$\phi_R(y,s) = \psi_R(|y|) + s^2/2$. Its axial gradient is unbounded. The usual bounded-gradient estimate cannot discard

$$2\,\|su\|_2\,\|\partial_s u\|_2\,.$$

An initially finite axial moment does not bound this factor uniformly in time. In our pair, the variance grows quadratically while the H^2 norm stays bounded. The companion audit identifies this closure issue explicitly. A proof gap by itself is not a counterexample; the independent existence and certificate arguments remain necessary.

The present work does not establish an open basin of negative-energy global data, a classification of all negative-energy dynamics, the same traveling-wave construction in every dimension, or a smooth cubic-type stable-manifold theory for the noninteger three-dimensional nonlinearity. Arbitrary higher-order dispersions and general nonlinearities remain possible extensions, not proved results of this manuscript.

9 Reproducibility and dependency map

The accompanying package contains the frozen inputs, interval verifiers, analytic proof notes and prior companion inputs required for the four-dimensional application. The following separation is maintained:

Conclusion	Necessary input
Three-dimensional traveling wave	exact profile residual; radial inverse; all even/odd angular channels; inverse pairing; analytic IFT and Pohozaev identity
Four-dimensional zero-momentum pair	previously certified negative-energy profile; compact metric lemma; moving-center estimates; weighted nonlinear contraction
Improved four-dimensional error radius	unchanged rational profile; mixed residual; original radial Ritz and HS estimates
Averaged spreading and grow-up	strong H^2 flow; conservation; interaction virial; mass transport; no numerical data

Rational polynomial data and binary floating witnesses are interpreted exactly before interval validation. Numerical eigenvalues are never accepted as a proof of an infinite-dimensional gap. Whole-space tails, origin regularity, compact-support matching, omitted spectral mass and transfer from q to Q are separate checks. A successful script verifies its stated inequalities, not every analytic assertion in the manuscript. Independent external replication of both parts remains a distinct step.

References

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